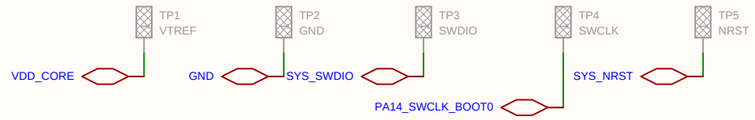
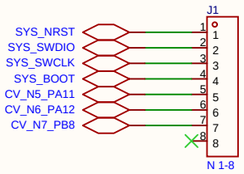
CraftVia Core-S G031 | v1 Rev0 - SCHEMATIC DRAFT

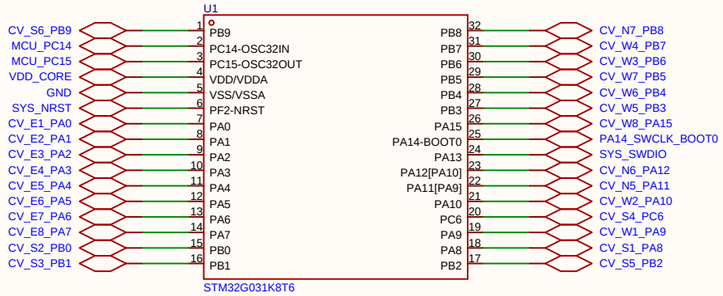
SWD rescue access (TP footprint provisional)



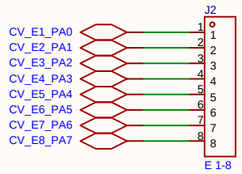
NORTH / J1.1-8 = N1-N8



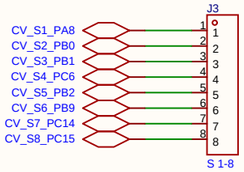
MCU / STM32G031K8T6



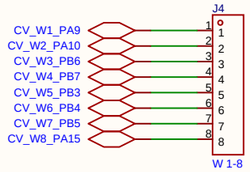
EAST / J2.1-8 = E1-E8



SOUTH / J3.1-8 = S1-S8



WEST / J4.1-8 = W1-W8



CLOCK OPTION - DESIGN NOTES

X1: ASEDVN-8.000MHZ-LR-T / 8 MHz CMOS.

Standard: internal HSI; external oscillator is optional.

Power OFF before changing links; match firmware to assembly.

External clock: HSE bypass, 8 MHz; disconnect S7 via R3.

Do not drive PC14 as GPIO with the external clock fitted.

Enable: R8 always ON, or R9 from PC15; never fit both.

Place X1 near PC14, C4 at X1 supply, R4 near X1 OUT.

Disable uses EN; supply remains connected. No LSE crystal.

Validate startup, clock switching, EN LOW and current on hardware.

POWER FILTER / RESET OPTIONS

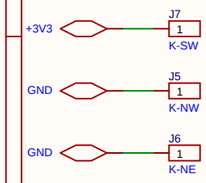


R7: FIT 0R by default; replace with qualified 0603 ferrite only.

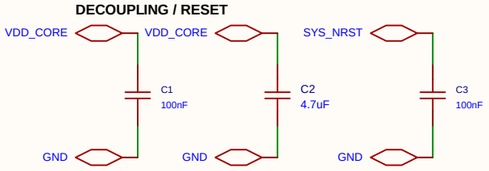
C1/C2, U1, TP1 and oscillator supply are downstream of R7.

Check bead DCR/current, voltage drop and LC resonance.

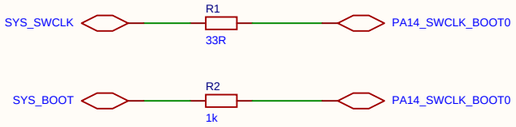
KEY / POWER



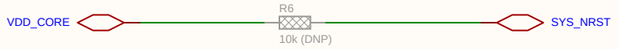
DECOUPLING / RESET



SWCLK / BOOT0 SHARED PIN



R1=33R and R2=1k: provisional; validate with Station.

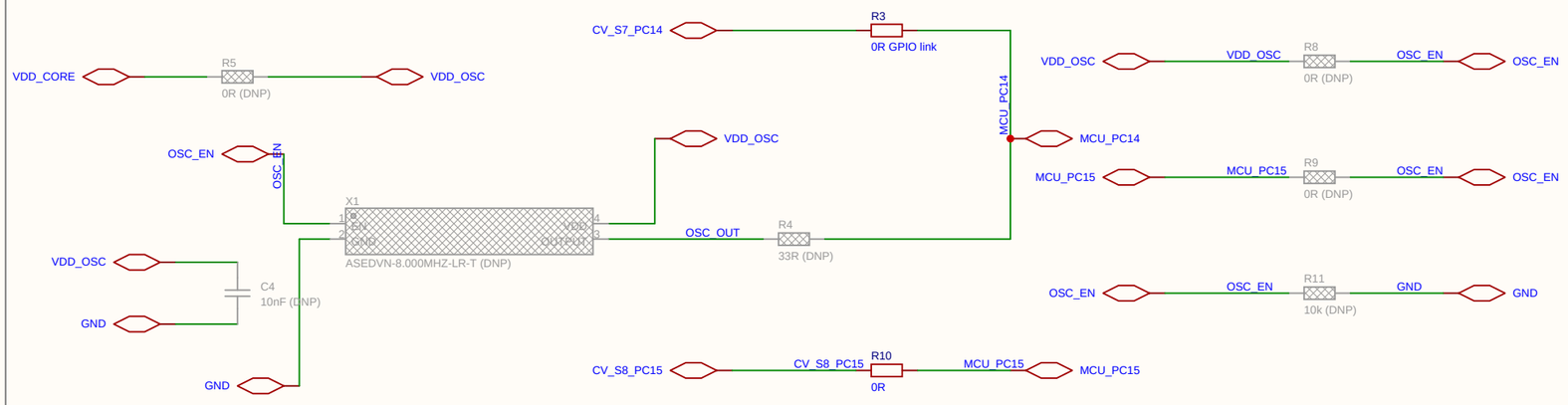


R6: optional NRST pull-up, 10k / 0603, DNP by default.

Internal pull-up + existing C3 remain. Validate reset timing.

No reverse-polarity / overvoltage protection provided by R7.

CLOCK SELECTION / ON-CORE OSCILLATOR OPTION



CLOCK ASSEMBLY - FIT LIST

HSI: R3,R10 FIT; clock option DNP

HSE always: X1,C4,R4,R5,R8,R10 FIT

HSE control: X1,C4,R4,R5,R9,R11 FIT

Never fit R8 and R9 together.

Control: remove R10; S8 isolated.

HSE: remove R3; S7 isolated.

PC15 AF1 = OSC_EN (or GPIO).

R11: EN pull-down; verify reset LOW.

X1 pin1=EN,2=GND,3=OUT,4=VDD

Dedicated ASEDVN footprint assigned; verify PCB orientation.

Lands 1.4x1.0mm; pitch 2.2x1.6mm

DESIGN NOTES - NOT RELEASED FOR FABRICATION

3.3 V input. Internal HSI standard; no regulator / USB / LED.
J1-J4: 2.54 mm headers; J5/J6=GND, J7=3V3.

Header height and TP bare-pad footprints: to be finalized.

SYS_BOOT: High-Z normally; disconnect SWCLK for BOOT.

R1/R2 do not provide SWCLK/BOOT electrical isolation.

BOOT option bytes and reset timing require validation.

C1/C2 close to U1 supply. PC14/15: restricted I/O drive.

Common contact positions: see Core-S series specification.

Schematic	Core-S-G031-Rev0	Create at	2026-09-03
		Update at	2026-09-29
Board	Board1	Page	Core-G031
Drawn		CraftVia-Core-S-G031-v1-Rev0	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A3	CraftVia,org